D4 Implementation Fidelity

Can the published formulation be verified in executable code?

Code availability

Public

Source code available

loss.py
model.py
⋮ `return` $\mathcal{L}_{phys}$

Not found

No accessible source code

code:
not available

Equation–code parity

If code is public, evaluate equation–code parity

Published equation

$\mathcal{L}_{\text{pub}}(\theta)$

Source code

$\mathcal{L}_{\text{exec}}(\theta)$

Aligned

Executable implementation matches formulation

$\mathcal{L}_{\text{pub}}(\theta) = \mathcal{L}_{\text{exec}}(\theta)$

Mismatch

Discrepancy between formulation and code

$\mathcal{L}_{\text{pub}}(\theta) \neq \mathcal{L}_{\text{exec}}(\theta)$